

The Odd-Fallback Collapse Theorem (m = 2 Family)

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Setup. We fix $r = 1$: squares fire on odd n , and the fallback rule applies on even n , making the allowed move from n to $n - a$. Let $G(n)$ denote the Grundy value of state n , with $G(0) = 0$.

Lemma A (P-positions = Even Numbers)

Theorem. For every odd integer $a \geq 1$, the set of P-positions (states n with $G(n) = 0$) consists precisely of all even integers. Consequently, the emergent period is 2 with a residue-set size of 1.

Proof. We proceed by induction on n .

Base case ($n = 0$). Terminal position, $G(0) = 0$, so 0 is a P-position.

Inductive step. Assume for all $k < n$ that $G(k) = 0 \iff k$ is even (equivalently, $G(k) > 0 \iff k$ is odd).

Case 1: n is even. The only available move is subtraction of the odd fallback a : $n \rightarrow n - a$. Since n is even and a is odd, the destination $n - a$ is strictly odd. By the induction hypothesis, $G(n - a) \neq 0$ (if $n - a \geq 0$), or the move set is empty (if $a > n$).

- If $a > n$: the move set is $\emptyset$, so $\text{mex}(\emptyset) = 0$ and $G(n) = 0$.
- If $a \leq n$: the set of reachable values is $\{G(n - a)\}$ with $G(n - a) \neq 0$, so $\text{mex}(\{G(n - a)\}) = 0$ and $G(n) = 0$.

In both cases $G(n) = 0$. Thus all even positions are P-positions.

Case 2: n is odd. The move set consists of square subtractions $n \rightarrow n - k^2$ for all integers $k \geq 1$ with $k^2 \leq n$. Choosing $k = 1$ gives a legal move to $n - 1$. Since n is odd, $n - 1$ is even, so $G(n - 1) = 0$ by the strong induction hypothesis. Thus $0 \in \{G(n - k^2) : k^2 \leq n\}$, forcing the mex to be positive, so $G(n) \neq 0$.

Therefore $G(n) = 0 \iff n \equiv 0 \pmod{2}$. ■

Lemma B (Full Sequence Equivalence)

Theorem. For all odd fallbacks $a, a' \geq 1$ and all states $n \geq 0$, the Grundy values satisfy $G_a(n) = G_{a'}(n)$.

Answers to the Referee Questions.

Question 1 (even positions). What is $\text{mex}(\{v\})$ when $v \neq 0$, and what does that force $G(n)$ to be for even n ?

For even n , the reachable set is either empty (if $n < a$) or $\{G(n - a)\}$. By Lemma A, since $n - a$ is odd, $G(n - a) = v > 0$. Now $\text{mex}(\emptyset) = 0$, and $\text{mex}(\{v\}) = 0$ whenever $v > 0$. Therefore $G(n) = 0$ for every even n , regardless of the specific value of v , and regardless of whether the move exists at all. The parameter a evaporates at every even step.

Question 2 (odd positions). At odd n , split the available squares k^2 by parity: which earlier positions' Grundy values enter the mex?

For odd n : if k is even, k^2 is even, so $n - k^2$ is odd; if k is odd, k^2 is odd, so $n - k^2$ is even. By Lemma A, for every odd k the landing $n - k^2$ is even with $G(n - k^2) = 0$. Thus the reachable Grundy values from odd n are $\{0\} \cup \{G(n - k^2) : k \text{ even}, k^2 \leq n\}$. The value 0 is guaranteed by the odd square moves ($k = 1, 3, 5, \dots$), and the non-zero entries depend exclusively on $G(n - k^2)$ for even k — landings on smaller odd numbers. So G at odd n is computed purely from earlier odd states, completely bypassing all even positions, and thus bypassing a entirely.

By induction on n : $G(0) = 0$; every even state yields $G = 0$ independent of a ; and every odd state's value is computed from prior odd states via even-square subtractions, in which a nowhere appears. The parameter a never impacts the Grundy sequence, so all odd fallbacks collapse to the exact same game. ■